

HARSH SINGH

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PROFESSIONAL SUMMARY

Senior Software Developer with **4+ years of experience** building enterprise backend applications using **Python, FastAPI, Django, and Flask**. Experienced in developing **REST APIs, microservices, SAP integrations, and cloud-based solutions** for Manufacturing Execution Systems (MES), Warehouse Management Systems (WMS), Traceability, Process Automation, Interlocking, and Packaging Solutions.

Skilled in **PostgreSQL, MySQL, Redis, RabbitMQ, Docker, Kubernetes, AWS, and CI/CD**. Hands-on experience integrating **OpenAI APIs and LangChain** to build AI-powered enterprise applications. Strong understanding of backend architecture, API design, database optimization, asynchronous processing, and Agile software development.

Skills

Programming Languages:	Python, SQL, JavaScript, HTML5, CSS3
Libraries/Frameworks:	FastAPI, Django, Flask, React.js, Pandas, OpenAI API, LangChain
Backend & Database:	REST APIs, Microservices, JWT, OpenAPI/Swagger, MS SQL Server, MySQL, MongoDB, Redis, RabbitMQ, Kafka, RAG
Tools & Deployment:	AWS (EC2, S3, RDS), Docker, Git, Linux, Postman, Jira, Pytest, CI/CD, Jenkins

PROFESSIONAL EXPERIENCE

Uno Minda Ltd. | Senior Software Developer

Gurgaon | July 2022 – Present

Enterprise Manufacturing Platform (MES & WMS)

- Developed backend services using Python, FastAPI, and Django.
- Built REST APIs for production, inventory, and warehouse operations.
- Integrated SAP, ERP, MES, and WMS systems.
- Designed modular microservices for business workflows.
- Optimized SQL queries and database performance.
- Implemented Redis caching to improve application performance.
- Used RabbitMQ for asynchronous task processing.
- Containerized applications using Docker.
- Deployed backend services on AWS.
- Supported CI/CD pipelines using GitHub Actions and Jenkins.
- Collaborated with QA, DevOps, and business teams in Agile environments.

Environment: Python, Django, PostgreSQL, Redis, RabbitMQ, Docker, AWS, SAP Integration, MongoDB

Traceability, Process Automation, Interlocking & Packaging Solutions

- Developed product traceability and genealogy modules.
- Built APIs for serial number and batch tracking.
- Developed manufacturing interlocking and validation logic.
- Automated production workflows and business processes.
- Built packaging validation and label management services.
- Integrated barcode and QR code workflows.
- Improved backend performance through query optimization and efficient API design.

Environment: Python, FastAPI, PostgreSQL, Redis, RabbitMQ, Docker, AWS, MongoDB

AI-Powered Manufacturing Knowledge Assistant

Python, FastAPI, OpenAI API, LangChain

- Developed an enterprise AI assistant using FastAPI, OpenAI APIs, and LangChain.
- Built APIs for document upload, semantic search, and knowledge retrieval.
- Implemented Retrieval-Augmented Generation (RAG).

KEY ACHIEVEMENTS

- Developed scalable backend applications for enterprise manufacturing solutions.
- Designed and maintained RESTful APIs for production and warehouse operations.
- Integrated SAP with manufacturing applications.
- Improved application performance through SQL optimization and Redis caching.
- Built containerized applications using Docker and AWS.
- Automated deployments through CI/CD pipelines.
- Developed AI-powered backend services using OpenAI APIs and LangChain.
- Participated in code reviews, production support, and performance optimization.

PROJECTS

Grocery Management System

Python | Flask | React.js | MS SQL

- Developed a full-stack Grocery Management System for inventory, supplier, and order management.
- Built REST APIs using Flask for product and inventory operations.
- Developed responsive user interfaces using React.js.
- Implemented authentication and database integration with MS SQL.

Handwritten Digit Recognition System

Python | TensorFlow | Keras | OpenCV

- Developed a handwritten digit recognition system using Convolutional Neural Networks (CNN).
- Trained the model on the MNIST dataset.
- Implemented image preprocessing and real-time digit prediction.
- Integrated the trained model into a Python application for inference.

EDUCATION

DIT University

Dehradun, India

Bachelor of Technology in Computer Science & Engineering

Aug 2018 - June 2022

Certifications

Machine Learning

- Coursera

Full Stack Development

- Code help

Internet of things (IOT)

- Udemy

ACHIEVEMENTS

- Awarded for Traceability Project at Uno Minda for factory automation.
- Saved ₹2 crore annually through EMS deployment.
- Led MES & IoT automation across multiple plants.
- Received Merit-Based Scholarship in college for academic excellence.